

A blueprint for the moduli stack of elliptic curves via fibered categories in Mathlib

Abstract

We give a detailed blueprint for the formalisation, in Lean 4/Mathlib, of the moduli stack of elliptic curves through the language of fibered categories. The construction proceeds in five parts. Part 1 records the Mathlib substrate. Part 2 fixes the base site (affine schemes with the fppf topology). Part 3 builds the fibered category $\mathcal{W} \rightarrow \text{Aff}$ of Weierstrass curves out of the existing `WeierstrassCurve` and `WeierstrassCurve.VariableChange` API, and proves that its projection is fibered in groupoids. Part 4 extracts the full subcategory $\mathcal{M}_{1,1}^{\mathcal{W}}$ of elliptic objects and records the quotient presentation $\mathcal{M}_{1,1}^{\mathcal{W}} \simeq [\text{Spec } A/G]$ with $A = \mathbb{Z}[a_1, a_2, a_3, a_4, a_6][\Delta^{-1}]$ and $G = \mathbb{G}_a^3 \rtimes \mathbb{G}_m$. Part 5 isolates the descent (stack) condition and reduces it to faithfully flat descent statements, several of which are not yet in Mathlib. Part 6 compares $\mathcal{M}_{1,1}^{\mathcal{W}}$ with the geometric stack $\mathcal{M}_{1,1}$ of proper smooth genus-one curves with a section and states the comparison theorem, whose hypotheses expose the scheme-theoretic prerequisites still missing from Mathlib. A milestone ladder (Part 7) orders the work by Mathlib-readiness.

Correctness note. Throughout, declarations already present in Mathlib are marked `[in Mathlib / formalised]` and annotated with their module path; all other items are targets. Every nontrivial target carries an explicit list of its prerequisites via `\uses`, so that the dependency graph is complete.

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1 Conventions and the Mathlib substrate

Convention 1.1. We work with Grothendieck universes as Mathlib does. A *category* means a term of `CategoryTheory.Category.{v,u}`; morphisms are written $X \rightarrow Y$, composition $f \gg g$ (“ f then g ”), identities 1_X . Opposite categories are $\mathcal{C}^{\text{op}}$. For a functor $p : \mathcal{X} \rightarrow \mathcal{S}$ we call $\mathcal{S}$ the *base* and $\mathcal{X}$ the *total category*.

Convention 1.2 (Rings and affine schemes). `CommRingCat` is the category of commutative rings (`Mathlib.Algebra.Category.Ring.Basic`). We set $\text{Aff} := \text{CommRingCat}^{\text{op}}$ and identify it with the category of affine schemes; an object of Aff is a ring R (written $\text{Spec } R$ when the geometric reading is intended), and a morphism $\text{Spec } R \rightarrow \text{Spec } R'$ in Aff is a ring homomorphism $f : R' \rightarrow R$, denoted $\underline{f}$ when regarded as an arrow of Aff . *Base change* along $\underline{f}$ is application of f .

Lean: `CommRingCat`

[in Mathlib / formalised]

1.1 The fibered-category API in Mathlib

The following are present in `Mathlib.CategoryTheory.FiberedCategory.*`; we recall the interface we will use.

Definition 1.3 (Lift of a morphism). For $p : \mathcal{X} \rightarrow \mathcal{S}$, objects $a, b \in \mathcal{X}$ over $R, S \in \mathcal{S}$ (i.e. $p(a) = R$, $p(b) = S$), a base arrow $f : R \rightarrow S$ and $\varphi : a \rightarrow b$, the proposition `p.IsHomLift f φ` asserts that $p(\varphi) = f$ modulo the equalities $p(a) = R$, $p(b) = S$ (implemented with `eqToHom`).

Lean: `CategoryTheory.Functor.IsHomLift`

[in Mathlib / formalised]

Definition 1.4 (Cartesian and strongly cartesian morphisms). With notation as in Definition 1.3 and φ a lift of $f : R \rightarrow S$:

- (a) φ is *cartesian* (`p.IsCartesian f φ`) if for every $a' \in \mathcal{X}$ over R and every $\psi : a' \rightarrow b$ lifting f there is a unique vertical $\chi : a' \rightarrow a$ (a lift of 1_R) with $\chi \gg \varphi = \psi$.
- (b) φ is *strongly cartesian* (`p.IsStronglyCartesian f φ`) if for every $a' \in \mathcal{X}$, every $g : p(a') \rightarrow R$ and every $\psi : a' \rightarrow b$ lifting $f \circ g$ there is a unique $\chi : a' \rightarrow a$ lifting g with $\chi \gg \varphi = \psi$.

Lean: `CategoryTheory.Functor.IsCartesian`, `CategoryTheory.Functor.IsStronglyCartesian`

[in Mathlib / formalised]

Lemma 1.5 (Basic API for (strongly) cartesian morphisms). *Strongly cartesian morphisms are stable under composition; a strongly cartesian morphism lifting an isomorphism is an isomorphism; two strongly cartesian lifts of the same arrow with the same target are related by a unique vertical isomorphism of their sources. All three are available in Mathlib.*

Lean: `CategoryTheory.Functor.IsStronglyCartesian.comp`, `CategoryTheory.Functor.IsStronglyCartesian.isIso_of_base.isIso`
`CategoryTheory.Functor.IsStronglyCartesian.domainUniqueUpToIso`
[in Mathlib / formalised]

Definition 1.6 (Fibered functor). $p : \mathcal{X} \rightarrow \mathcal{S}$ is *prefibered* (`p.IsPreFibered`) if for every $f : R \rightarrow S$ in $\mathcal{S}$ and every b over S there is a strongly cartesian $\varphi : a \rightarrow b$ lifting f ; it is *fibered* (`p.IsFibered`) if in addition strongly cartesian morphisms are stable under composition (automatic from Lemma 1.5). Mathlib packages the prefibered pullback as `pullbackObj` and `pullbackMap`.

Lean: `CategoryTheory.Functor.IsPreFibered`, `CategoryTheory.Functor.IsFibered`, `CategoryTheory.Functor.pullbackObj`,
`CategoryTheory.Functor.pullbackMap`
[in Mathlib / formalised]

Definition 1.7 (Fibre categories). For $S \in \mathcal{S}$, $p.\text{Fiber } S$ is the subtype $\{a : \mathcal{X} \mid p(a) = S\}$, made a category with morphisms the vertical morphisms (lifts of 1_S). The inclusion `fiberInclusion` into $\mathcal{X}$ is faithful, and every φ in the fibre satisfies `p.IsHomLift 1_S` φ .

Lean: `CategoryTheory.Functor.Fiber`, `CategoryTheory.Functor.Fiber.fiberInclusion`
[in Mathlib / formalised]

Definition 1.8 (Fibered in groupoids). A fibered functor p is *fibered in groupoids* (a *CFG*) when every morphism of $\mathcal{X}$ is strongly cartesian; equivalently every fibre $p.\text{Fiber } S$ is a groupoid and p is prefibered. The equivalence of the two formulations is standard [3, Def. 3.21, §3.6] and is the target lemma `isFibered_iff_fibers_groupoid` below (see Proposition 3.32).

Remark 1.9 (The Grothendieck and CoGrothendieck constructions). Mathlib (`Mathlib.CategoryTheory.Bicategory`) provides two Grothendieck constructions for a pseudofunctor valued in **Cat** [7]. For a covariant $F : \text{LocallyDiscrete } \mathcal{S} \rightarrow \mathbf{Cat}$ it provides $\int F$ (`Pseudofunctor.Grothendieck`), whose objects are pairs (S, a) with $a \in F(S)$ and whose morphisms $(R, b) \rightarrow (S, a)$ are pairs (f, h) with $f : R \rightarrow S$ in $\mathcal{S}$ and $h : F(f)(a) \rightarrow b$. For a contravariant $F : \text{LocallyDiscrete } \mathcal{S}^{\text{op}} \rightarrow \mathbf{Cat}$ it provides $\int^c F$ (`Pseudofunctor.CoGrothendieck`), whose objects are pairs (S, a) with $a \in F(S)$ and whose morphisms $(R, b) \rightarrow (S, a)$ are pairs (f, h) with $f : R \rightarrow S$ in $\mathcal{S}$ and $h : b \rightarrow F(f)(a)$, together with the projection `forget F` : $\int^c F \rightarrow \mathcal{S}$, $(S, a) \mapsto S$, $(f, h) \mapsto f$. The Mathlib naming convention reserves *Grothendieck* for the covariant construction and *CoGrothendieck* for the contravariant one, consistent with `CategoryTheory.Grothendieck` for 1-functors. The moduli problem is contravariant in $\mathcal{S} = \mathbf{Aff}$, so the relevant construction is $\int^c F$; see Construction 3.35.

Lean: `CategoryTheory.Pseudofunctor.Grothendieck`, `CategoryTheory.Pseudofunctor.CoGrothendieck`
[in Mathlib / formalised]

Proposition 1.10 (The CoGrothendieck construction is a fibered category). *For $F : \text{LocallyDiscrete } \mathcal{S}^{\text{op}} \rightarrow \mathbf{Cat}$ the projection `forget F` : $\int^c F \rightarrow \mathcal{S}$ is a fibered category:*

- (a) *for $f : R \rightarrow S$ in $\mathcal{S}$ and $a \in F(S)$, the arrow `cartesianLift a f` with domain $(R, F(f)(a))$ is strongly cartesian;*
- (b) *`forget F` carries an `IsFibered` instance;*
- (c) *`forget F` carries a `HasFibers` instance whose fibre over S is $F(S)$, and the induced functor $F(S) \rightarrow (\text{forget } F).\text{Fiber } S$ is an equivalence.*

All three are established in `Mathlib.CategoryTheory.FiberedCategory.Grothendieck` (using `FiberedCategory.HasFibers`).

Lean: `CategoryTheory.Pseudofunctor.CoGrothendieck.isStronglyCartesian_homCartesianLift`, `CategoryTheory.Pseudofunctor.CoGrothendieck.instHasFibersForget`
[in Mathlib / formalised]

1.2 Sites and sheaves in Mathlib

Definition 1.11 (Grothendieck topology, sheaf). Mathlib has sieves, Grothendieck topologies (maximality, stability under pullback, transitivity), pretopologies, coverages, the sheaf condition for presheaves valued in an arbitrary category, sheafification for suitable targets, and the comparison lemma between pretopologies and topologies.

Lean: `CategoryTheory.GrothendieckTopology`, `CategoryTheory.Pretopology`, `CategoryTheory.Sieve`, `CategoryTheory.Presheaf`, `CategoryTheory.Sheaf`

[in Mathlib / formalised]

Warning 1.12 (Stacks are not in Mathlib). As of the reference commit, Mathlib contains *sheaf* theory (a 1-truncated descent theory) but no notion of *stack* (descent for a fibered category), no stackification, and no fppf/étale topology on schemes as a named object. All of Part 5 that concerns descent *of objects* is therefore new content. This is the principal gap between the present blueprint and a finished formalisation.

1.3 The elliptic-curve API in Mathlib

Definition 1.13 (Weierstrass curve). For $R : \text{CommRing}$, `WeierstrassCurve` R is the five-tuple $(a_1, a_2, a_3, a_4, a_6) \in R^5$. Derived quantities $b_2, b_4, b_6, b_8, c_4, c_6$, and the discriminant Δ are defined, with $c_4^3 - c_6^2 = 1728 \Delta$ and $\Delta = (-b_2^2 b_8 - 8b_4^3 - 27b_6^2 + 9b_2 b_4 b_6)$. Base change along $f : R \rightarrow A$ is $W.\text{map } f : \text{WeierstrassCurve } A$, obtained by applying f to each coefficient.

Lean: `WeierstrassCurve`, `WeierstrassCurve.Delta`, `WeierstrassCurve.map`

[in Mathlib / formalised]

Lemma 1.14 (Functoriality of base change). $W.\text{map } (\text{RingHom.id } R) = W$ and $(W.\text{map } f).\text{map } g = W.\text{map } (g \circ f)$; moreover $(W.\text{map } f).\Delta = f(W.\Delta)$ and likewise for b_i, c_4, c_6 . Injectivity of $W \mapsto W.\text{map } f$ for injective f is *map_injective*.

Lean: `WeierstrassCurve.map_id`, `WeierstrassCurve.map_map`, `WeierstrassCurve.map_injective`

[in Mathlib / formalised]

Definition 1.15 (Variable changes). `WeierstrassCurve.VariableChange` R is the datum $C = (u, r, s, t)$ with $u \in R^\times$ and $r, s, t \in R$. These form a group under the classical composition law [6, III, Table 3.1], and act on `WeierstrassCurve` R by

$$C \cdot W = (a'_1, a'_2, a'_3, a'_4, a'_6),$$

where, writing $u^{-1} \in R^\times$,

$$\begin{aligned} a'_1 &= u^{-1}(a_1 + 2s), & a'_2 &= u^{-2}(a_2 - sa_1 + 3r - s^2), \\ a'_3 &= u^{-3}(a_3 + ra_1 + 2t), & a'_4 &= u^{-4}(a_4 - sa_3 + 2ra_2 - (t + rs)a_1 + 3r^2 - 2st), \\ a'_6 &= u^{-6}(a_6 + ra_4 + r^2a_2 + r^3 - ta_3 - t^2 - rta_1). \end{aligned}$$

This is a left action ($\mathbf{1} \cdot W = W$, $(C_1 C_2) \cdot W = C_1 \cdot (C_2 \cdot W)$).

Lean: `WeierstrassCurve.VariableChange`, `WeierstrassCurve.instGroupVariableChange`, `WeierstrassCurve.instMulActionVariableChange`

[in Mathlib / formalised]

Lemma 1.16 (Transformation of invariants). For $C = (u, r, s, t)$ one has $(C \cdot W).\Delta = u^{-12} W.\Delta$, $(C \cdot W).c_4 = u^{-4} W.c_4$, $(C \cdot W).c_6 = u^{-6} W.c_6$.

Lean: `WeierstrassCurve.variableChange_Delta`, `WeierstrassCurve.variableChange_c4`, `WeierstrassCurve.variableChange_c6`

[in Mathlib / formalised]

Definition 1.17 (Elliptic curve; j -invariant). `WeierstrassCurve.IsElliptic` W is the class asserting that Δ is a unit; under it $\Delta' \in R^\times$ is the unit with $(\Delta' : R) = \Delta$, and the j -invariant is $j := (\Delta')^{-1}c_4^3$. Every $j_0 \in F$ over a field F is realised by an explicit curve `ofJ j0`.

Lean: `WeierstrassCurve.IsElliptic`, `WeierstrassCurve.Delta'`, `WeierstrassCurve.coe.Delta'`, `WeierstrassCurve.j`, `WeierstrassCurve.ofJ`

[in Mathlib / formalised]

Warning 1.18 (The affine model is a presentation, not $\mathcal{M}_{1,1}$). The Mathlib module `docstring` records the decisive caveat: the type `WeierstrassCurve R` with `IsElliptic` “gives a type which can be beefed up to a category which is equivalent to the category of elliptic curves over $\text{Spec } R$ in the case that R has trivial Picard group $\text{Pic}(R)$ or, slightly more generally, when its 12-torsion is trivial.” In general the two differ by line-bundle twists; see Part 6. The object built in Parts 3–4 is therefore the *Weierstrass presentation* of $\mathcal{M}_{1,1}$, whose fppf-stackification is $\mathcal{M}_{1,1}$.

2 The base site: affine schemes with the fppf topology

Definition 2.1 (fppf covers of an affine scheme). A *standard fppf cover* of $R \in \text{Aff}$ is a finite family of ring maps $\{R \rightarrow R_i\}_{i \in I}$, each flat and of finite presentation, such that $R \rightarrow \prod_i R_i$ is faithfully flat (equivalently $\text{Spec}(\prod_i R_i) \rightarrow \text{Spec } R$ is surjective). The associated sieve on $\text{Spec } R$ in Aff is generated by the $(R \rightarrow R_i) : \text{Spec } R_i \rightarrow \text{Spec } R$.

Lean:

[not ready]

Lemma 2.2 (The standard fppf data form a pretopology). *The families of Definition 2.1 contain the isomorphisms, are stable under base change (pullback of a flat finite-presentation map is again such, and faithful flatness is preserved), and are stable under composition (refinement). Hence they define a Pretopology on Aff, whose associated Grothendieck topology is the fppf topology τ_{fppf} .*

Lean:

[not ready]

Proof sketch. Pullback in Aff is the tensor product: the base change of $R \rightarrow R_i$ along $R \rightarrow R'$ is $R' \rightarrow R' \otimes_R R_i$. Flatness and finite presentation are stable under base change (`Module.Flat.baseChange`, `Algebra.FinitePresentation.base-change` lemmas). Faithful flatness of $R \rightarrow \prod_i R_i$ pulls back to faithful flatness of $R' \rightarrow \prod_i R' \otimes_R R_i$ (`Module.FaithfullyFlat.base-change`). Stability under composition uses that a composite of flat (resp. finitely presented, resp. faithfully flat) maps is again such. The pretopology axioms of `CategoryTheory.Pretopology` then hold by construction. The Zariski and étale (co)topologies arise by restricting to open immersions (localisations $R \rightarrow R_f$) resp. to étale maps; they are coarser than τ_{fppf} . $\square$

Remark 2.3 (Set-theoretic size). To stay within Mathlib’s universe discipline one works either with the *small* fppf site of a fixed base ring R_0 (the slice $\text{Aff}_{/\text{Spec } R_0}$ of finitely presented R_0 -algebras) or with a suitable small category of rings. The moduli constructions below are insensitive to this choice; we write Aff generically and fix a small model when formalising.

3 The fibered category of Weierstrass curves

Throughout this section base arrows are read as in Convention 1.2: an Aff-arrow $\underline{f} : \text{Spec } R \rightarrow \text{Spec } R'$ is a ring map $f : R' \rightarrow R$.

3.1 Equivariance of the action under base change

Definition 3.1 (Base change of a variable change). For a ring homomorphism $f : R \rightarrow A$ define $\text{map } f : \text{VC}(R) \rightarrow \text{VC}(A)$ on underlying data by $(u, r, s, t) \mapsto (\text{Units.map } f \, u, f(r), f(s), f(t))$.

Lean: `WeierstrassCurve.VariableChange.map`

[not ready]

Lemma 3.2 ($\text{map } f$ preserves the identity). For $f : R \rightarrow A$, $\text{map } f \, \mathbf{1} = \mathbf{1}$ in $\text{VC}(A)$.

Lean: `WeierstrassCurve.VariableChange.map_one`

[not ready]

Lemma 3.3 ($\text{map } f$ preserves the product). For $f : R \rightarrow A$ and $C_1, C_2 \in \text{VC}(R)$, $\text{map } f \, (C_1 C_2) = (\text{map } f \, C_1)(\text{map } f \, C_2)$; hence $\text{map } f$ is a group homomorphism $\text{VC}(R) \rightarrow \text{VC}(A)$.

Lean: `WeierstrassCurve.VariableChange.map_mul`

[not ready]

Lemma 3.4 (Functoriality of map : identity). $\text{map}(\text{RingHom.id } R) = \text{id}_{\text{VC}(R)}$.

Lean: `WeierstrassCurve.VariableChange.map_id`

[not ready]

Lemma 3.5 (Functoriality of map : composition). For $f : R \rightarrow A$ and $g : A \rightarrow B$ one has $\text{map}(g \circ f) = \text{map } g \circ \text{map } f$; equivalently $\text{map } g \, (\text{map } f \, C) = \text{map}(g \circ f) \, C$ for all $C \in \text{VC}(R)$.

Lean: `WeierstrassCurve.VariableChange.map_map`

[not ready]

Lemma 3.6 (Equivariance of the action under base change). For $f : R \rightarrow A$, $C \in \text{VC}(R)$ and $W : \text{WeierstrassCurve } R$,

$$(C \cdot W).\text{map } f = (\text{map } f \, C) \cdot (W.\text{map } f).$$

Lean: `WeierstrassCurve.map_variableChange`

[not ready]

Proof sketch. Both sides are Weierstrass curves over A ; by `WeierstrassCurve.ext` it suffices to compare the five coefficients $a_1, \dots, a_6$. Each coefficient of the left side is f applied to the corresponding coefficient of $C \cdot W$, i.e. f applied to a polynomial in u^{-1}, r, s, t, a_i (Definition 1.15); each coefficient of the right side is the same polynomial evaluated at $\text{Units.map } f \, u^{-1} = f(u^{-1})$, $f(r), f(s), f(t), f(a_i)$. The two agree because f is a ring homomorphism and $\text{Units.map } f$ commutes with inversion; each coefficient identity is closed by `map_mul`, `map_add`, `map_pow`, `Units.coe_map` and `ring`. $\square$

3.2 The total category

Definition 3.7 (Objects of $\mathcal{W}$). An object of $\mathcal{W}$ is a pair (R, W) with $R \in \text{Aff}$ and $W : \text{WeierstrassCurve } R$.

Lean:

[not ready]

Definition 3.8 (Morphisms of $\mathcal{W}$). A morphism $(R, W) \rightarrow (R', W')$ is a pair $(\underline{f}, C)$ where $\underline{f} : \text{Spec } R \rightarrow \text{Spec } R'$ is an Aff-arrow (a ring map $f : R' \rightarrow R$) and $C \in \text{VC}(R)$ satisfies

$$C \cdot W = W'.\text{map } f. \quad (1)$$

Two morphisms are equal iff their Aff-arrows and their VC-components agree, the condition (1) being a proposition.

Lean:

[not ready]

Lemma 3.9 (The identity datum is a morphism). *For $(R, W) \in \mathcal{W}$ the pair $(\text{id}_R, \mathbf{1})$ satisfies (1): $\mathbf{1} \cdot W = W = W.\text{map}(\text{RingHom.id } R)$.*

Lean:

[not ready]

Definition 3.10 (Identity morphism). The identity of (R, W) is $\mathbf{1}_{(R, W)} := (\text{id}_R, \mathbf{1})$, well-defined by Lemma 3.9.

Lean:

[not ready]

Lemma 3.11 (The composite datum is a morphism). *Given $(R, W) \xrightarrow{(\underline{f}, C)} (R', W') \xrightarrow{(\underline{g}, C')} (R'', W'')$ (so $f : R' \rightarrow R$, $g : R'' \rightarrow R'$), the pair $(\underline{f \circ g}, (\text{map } f C') \cdot C)$ satisfies (1) over R :*

$$((\text{map } f C') \cdot C) \cdot W = W''.\text{map } (f \circ g).$$

Lean:

[not ready]

Proof. Using $C \cdot W = W'.\text{map } f$ and $C' \cdot W' = W''.\text{map } g$,

$$\begin{aligned} ((\text{map } f C') \cdot C) \cdot W &= (\text{map } f C') \cdot (C \cdot W) && (\text{mul_smul}) \\ &= (\text{map } f C') \cdot (W'.\text{map } f) && (\text{hypothesis on } C) \\ &= (C' \cdot W').\text{map } f && (\text{Lemma 3.6}) \\ &= (W''.\text{map } g).\text{map } f && (\text{hypothesis on } C') \\ &= W''.\text{map } (f \circ g) && (\text{Lemma 1.14}). \end{aligned}$$

□

Definition 3.12 (Composition). The composite of $(\underline{f}, C)$ and $(\underline{g}, C')$ as above is

$$(\underline{f}, C) \gg (\underline{g}, C') := (\underline{f \circ g}, (\text{map } f C') \cdot C), \quad (2)$$

well-defined by Lemma 3.11.

Lean:

[not ready]

Lemma 3.13 (Left identity). $\mathbf{1}_{(R, W)} \gg (\underline{f}, C) = (\underline{f}, C)$ for every $(\underline{f}, C) : (R, W) \rightarrow (R', W')$.

Lean:

[not ready]

Proof. The Aff-component is $\underline{\text{id}}_R \circ f = \underline{f}$; the VC-component is $(\text{map id } C) \cdot \mathbf{1} = C \cdot \mathbf{1} = C$, using Lemma 3.4 and `mul_one`. $\square$

Lemma 3.14 (Right identity). $(\underline{f}, C) \gg \mathbf{1}_{(R', W')} = (\underline{f}, C)$.

Lean:

[not ready]

Proof. The Aff-component is $\underline{f} \circ \underline{\text{id}}_{R'} = \underline{f}$; the VC-component is $(\text{map } f \mathbf{1}) \cdot C = \mathbf{1} \cdot C = C$, using Lemma 3.2 and `one_mul`. $\square$

Lemma 3.15 (Associativity). *For composable $(\underline{f}, C), (\underline{g}, C'), (\underline{h}, C'')$,*

$$((\underline{f}, C) \gg (\underline{g}, C')) \gg (\underline{h}, C'') = (\underline{f}, C) \gg ((\underline{g}, C') \gg (\underline{h}, C'')).$$

Lean:

[not ready]

Proof. Both Aff-components equal $\underline{f} \circ \underline{g} \circ \underline{h}$. The VC-components are

$$(\text{map } (f \circ g) C'') \cdot ((\text{map } f C') \cdot C) \quad \text{and} \quad (\text{map } f ((\text{map } g C'') \cdot C')) \cdot C.$$

Expanding the second by Lemma 3.3 (map f preserves products) and Lemma 3.5 (map f (map $g C''$) = map $(f \circ g) C''$) yields $((\text{map } (f \circ g) C'') \cdot (\text{map } f C')) \cdot C$, equal to the first by `mul_assoc`. $\square$

Definition 3.16 (The category $\mathcal{W}$). The objects (Definition 3.7), morphisms (Definition 3.8), identities (Definition 3.10) and composition (Definition 3.12), together with Lemmas 3.13, 3.14 and 3.15, constitute a category $\mathcal{W}$.

Lean: `Mathlib.AlgebraicGeometry.EllipticCurve.Moduli.Weierstrass`

[not ready]

Definition 3.17 (The projection $p_{\mathcal{W}}$). $p_{\mathcal{W}} : \mathcal{W} \rightarrow \text{Aff}$ sends $(R, W) \mapsto \text{Spec } R$ and $(\underline{f}, C) \mapsto \underline{f}$. It preserves identities and composition by Definitions 3.10 and 3.12, whose Aff-components are $\underline{\text{id}}$ and the composite of the Aff-components respectively.

Lean:

[not ready]

3.3 Cartesian structure

Lemma 3.18 (Morphisms are homlifts). *Every morphism $(\underline{f}, C) : (R, W) \rightarrow (R', W')$ satisfies $p_{\mathcal{W}}.\text{IsHomLift } \underline{f}(\underline{f}, C)$, by definition of the projection.*

Lean:

[not ready]

Lemma 3.19 (Vertical morphisms). *A morphism $(\underline{g}, C) : (R, W) \rightarrow (R, W_0)$ is vertical (a lift of $\mathbf{1}_R$) iff $g = \text{id}_R$; the vertical morphisms $(R, W) \rightarrow (R, W_0)$ are then exactly the $C \in \text{VC}(R)$ with $C \cdot W = W_0$.*

Lean:

[not ready]

Lemma 3.20 (Fibre hom-sets). *For $R \in \text{Aff}$ and W, W_0 over R , the hom-set in the fibre $p_{\mathcal{W}}.\text{Fiber}(\text{Spec } R)$ is $\text{Hom}(W, W_0) = \{C \in \text{VC}(R) \mid C \cdot W = W_0\}$.*

Lean:

[not ready]

Lemma 3.21 (Fibre composition is the group law). *Composition in $p_{\mathcal{W}}.\text{Fiber}(\text{Spec } R)$ is the group law of $\text{VC}(R)$: the composite of $C : W \rightarrow W_0$ and $C' : W_0 \rightarrow W_1$ has VC-component $C' \cdot C$.*

Lean:

[not ready]

Proof. By (2) with $f = g = \text{id}_R$ the composite has VC-component $(\text{map id } C') \cdot C = C' \cdot C$, using Lemma 3.4. $\square$

Lemma 3.22 (The fibre is a groupoid). *$p_{\mathcal{W}}.\text{Fiber}(\text{Spec } R)$ is a groupoid: a morphism $C : W \rightarrow W_0$ (so $C \cdot W = W_0$) has two-sided inverse $C^{-1} : W_0 \rightarrow W$, where $C^{-1} \cdot W_0 = W$.*

Lean:

[not ready]

Proof. $C^{-1} \cdot W_0 = C^{-1} \cdot (C \cdot W) = (C^{-1}C) \cdot W = W$ by `mul_smul` and `inv_mul_cancel`; the two composites are $C^{-1}C = \mathbf{1}$ and $CC^{-1} = \mathbf{1}$ (Lemma 3.21). $\square$

Lemma 3.23 (The fibre is the translation groupoid). *For $R \in \text{Aff}$ the fibre $p_{\mathcal{W}}.\text{Fiber}(\text{Spec } R)$ is isomorphic, as a category, to the action groupoid $\text{VC}(R) \ltimes \text{WeierstrassCurve } R$ (objects Weierstrass curves over R ; $\text{Hom}(W, W_0) = \{C \mid C \cdot W = W_0\}$; composition the group law). The isomorphism is the identity on objects and $C \mapsto C$ on morphisms.*

Lean:

[not ready]

Lemma 3.24 (Base-change data is a morphism). *For $\underline{f} : \text{Spec } R \rightarrow \text{Spec } R'$ (ring map $f : R' \rightarrow R$) and $(R', W') \in \mathcal{W}$, the pair $\lambda_{f, W'} := (\underline{f}, \mathbf{1})$ is a morphism $(R, W'.\text{map } f) \rightarrow (R', W')$: indeed $\mathbf{1} \cdot (W'.\text{map } f) = W'.\text{map } f$.*

Lean:

[not ready]

Lemma 3.25 (Existence of the cartesian factorisation). *Let $(A, V) \in \mathcal{W}$, let $\underline{h} : \text{Spec } A \rightarrow \text{Spec } R$ (ring map $h : R \rightarrow A$), and let $\psi = (\underline{k}, D) : (A, V) \rightarrow (R', W')$ lift $\underline{f} \circ \underline{h}$, so $k = h \circ f$ and $D \cdot V = W'.\text{map } k$. Then $\chi := (\underline{h}, D)$ is a morphism $(A, V) \rightarrow (R, W'.\text{map } f)$ and $\chi \gg \lambda_{f, W'} = \psi$.*

Lean:

[not ready]

Proof. Legitimacy of χ is $D \cdot V = (W'.\text{map } f).\text{map } h$, which holds because $D \cdot V = W'.\text{map } (h \circ f) = (W'.\text{map } f).\text{map } h$ by Lemma 1.14. By (2), $\chi \gg \lambda_{f, W'}$ has Aff-component $\underline{h} \circ \underline{f} = \underline{k}$ and VC-component $(\text{map } h \mathbf{1}) \cdot D = D$, so $\chi \gg \lambda_{f, W'} = \psi$. $\square$

Lemma 3.26 (Uniqueness of the cartesian factorisation). *With notation as in Lemma 3.25, any morphism $\chi' = (\underline{h}, E) : (A, V) \rightarrow (R, W'.\text{map } f)$ with $\chi' \gg \lambda_{f, W'} = \psi$ has $E = D$; hence $\chi' = \chi$.*

Lean:

[not ready]

Proof. By (2) the VC-component of $\chi' \gg \lambda_{f,W'}$ is $(\text{map } h \mathbf{1}) \cdot E = E$; equating with that of ψ gives $E = D$. $\square$

Lemma 3.27 (Base-change arrows are strongly cartesian). *For $\underline{f} : \text{Spec } R \rightarrow \text{Spec } R'$ and $(R', W') \in \mathcal{W}$, the arrow $\lambda_{f,W'} = (\underline{f}, \mathbf{1}) : (R, W'.\text{map } f) \rightarrow (R', W')$ is strongly cartesian.*

Lean:

[not ready]

Proof. Lemmas 3.25 and 3.26 provide the unique factorisation required by Definition 1.4(b). $\square$

Lemma 3.28 ($p_{\mathcal{W}}$ is prefibered). *$p_{\mathcal{W}}$ is prefibered: for every $\underline{f} : \text{Spec } R \rightarrow \text{Spec } R'$ and $(R', W') \in \mathcal{W}$, the arrow $\lambda_{f,W'}$ (Lemma 3.27) is a strongly cartesian lift of $\underline{f}$ with target (R', W') .*

Lean:

[not ready]

Theorem 3.29 ($p_{\mathcal{W}}$ is fibered). *$p_{\mathcal{W}} : \mathcal{W} \rightarrow \text{Aff}$ is a fibered category (IsFibered).*

Lean:

[not ready]

Proof. $p_{\mathcal{W}}$ is prefibered (Lemma 3.28) and strongly cartesian morphisms compose (Lemma 1.5). $\square$

Proposition 3.30 (Groupoid fibres give a CFG). *Let $p : \mathcal{X} \rightarrow \mathcal{S}$ be prefibered. If every fibre $p.\text{Fiber } S$ is a groupoid, then every morphism of $\mathcal{X}$ is strongly cartesian; i.e. p is fibered in groupoids.*

Lean:

[not ready]

Proof sketch. Given $\varphi : a \rightarrow b$ over $f : R \rightarrow S$ and a test $\psi : a' \rightarrow b$ over $f \circ g$, let $\lambda : f^*b \rightarrow b$ be a strongly cartesian lift (prefiberedness). Then $\psi = \chi_0 \gg \lambda$ with χ_0 over g , and $\varphi = v \gg \lambda$ with v vertical, hence invertible as the fibre is a groupoid; $\chi := \chi_0 \gg v^{-1}$ is the unique factorisation, so φ is strongly cartesian. $\square$

Proposition 3.31 (A CFG has groupoid fibres). *If $p : \mathcal{X} \rightarrow \mathcal{S}$ is fibered in groupoids, then every fibre is a groupoid.*

Lean:

[not ready]

Proof sketch. A fibre morphism is vertical, hence lifts the isomorphism $\mathbf{1}_S$; being strongly cartesian it is an isomorphism (Lemma 1.5). $\square$

Proposition 3.32 (Groupoid criterion). *A prefibered $p : \mathcal{X} \rightarrow \mathcal{S}$ is fibered in groupoids iff every fibre is a groupoid.*

Lean:

[not ready]

Corollary 3.33 ($\mathcal{W}$ is a CFG). *$p_{\mathcal{W}}$ is fibered in groupoids: it is prefibered (Lemma 3.28) and each fibre is a groupoid (Lemma 3.22); apply Proposition 3.30.*

Lean:

[not ready]

3.4 The pseudofunctor presentation (optional route)

Definition 3.34 (The base-change functor). For a ring map $f : R \rightarrow A$, the assignment $W \mapsto W.\text{map } f$ on objects and $C \mapsto \text{map } f C$ on morphisms is a functor $\mathbf{F}(f) : \mathbf{F}(R) \rightarrow \mathbf{F}(A)$ between the fibre groupoids of Lemma 3.23, where $\mathbf{F}(R) := \text{VC}(R) \ltimes \text{WeierstrassCurve } R$; functoriality on morphisms is Lemma 3.6.

Lean:

[not ready]

Construction 3.35 (The pseudofunctor $\mathbf{F}$). The data $R \mapsto \mathbf{F}(R)$, $f \mapsto \mathbf{F}(f)$ (Definition 3.34) assemble into a pseudofunctor $\mathbf{F} : \text{LocallyDiscrete CommRingCat} \rightarrow \text{Cat}$: the unitor $\mathbf{F}(\text{id}) \cong \text{id}$ and the associator $\mathbf{F}(g) \circ \mathbf{F}(f) \cong \mathbf{F}(g \circ f)$ are the identity 2-cells supplied by `map_id` and `map_map` (Lemma 1.14), and the pentagon and triangle coherences hold because these 2-cells are identities.

Lean:

[not ready]

Construction 3.36 ($\mathcal{W}$ as a CoGrothendieck construction). Under the identification $\text{Aff}^{\text{op}} = \text{CommRingCat}$, $\mathbf{F}$ is a contravariant pseudofunctor on Aff , and its CoGrothendieck construction (Remark 1.9) satisfies $\int^c \mathbf{F} \cong \mathcal{W}$ over Aff , with the Mathlib projection `forget  $\mathbf{F} : \int^c \mathbf{F} \rightarrow \text{Aff}$` equal to $p_{\mathcal{W}}$ (Definition 3.17) across this isomorphism.

Lean: `CategoryTheory.Pseudofunctor.CoGrothendieck`, `CategoryTheory.Pseudofunctor.CoGrothendieck.forget`

[not ready]

Remark 3.37. Along this route the fibered-category conclusions are supplied by Mathlib: by Proposition 1.10, `forget  $\mathbf{F}$` carries an `IsFibered` instance (with strongly cartesian lifts `cartesianLift`), which is Theorem 3.29 and the prefibered part of Corollary 3.33 transported across $\int^c \mathbf{F} \cong \mathcal{W}$; since each fibre $\mathbf{F}(R)$ is a groupoid (Lemma 3.22), Proposition 3.30 gives Corollary 3.33. The obligation this route incurs is Construction 3.35, i.e. producing the pseudofunctor $\mathbf{F}$ with its coherences. The direct route (Definitions 3.7–Corollary 3.33) instead proves strong cartesianness by hand (Lemma 3.27) and does not require the bicategorical coherences.

4 The moduli stack of elliptic curves (Weierstrass model)

4.1 The moduli category and its fibered structure

Lemma 4.1 (`IsElliptic` is stable under base change). *If $W.\text{IsElliptic}$ and $f : R \rightarrow A$ then $(W.\text{map } f).\text{IsElliptic}$. Indeed $(W.\text{map } f).\Delta = f(W.\Delta)$ (Lemma 1.14) is a unit, since a ring homomorphism sends units to units.*

Lean: `WeierstrassCurve.IsElliptic.map`

[not ready]

Lemma 4.2 (`IsElliptic` is stable under variable change). *If $W.\text{IsElliptic}$ and $C = (u, r, s, t) \in \text{VC}(R)$ then $(C \cdot W).\text{IsElliptic}$. Indeed $(C \cdot W).\Delta = u^{-12} W.\Delta$ (Lemma 1.16) is a product of units.*

Lean:

[not ready]

Definition 4.3 (The Weierstrass moduli category). $\mathcal{M}_{1,1}^W$ is the full subcategory of $\mathcal{W}$ on the objects (R, W) with $W.\text{IsElliptic}$.

Lean:

[not ready]

Definition 4.4 (The moduli projection). $p : \mathcal{M}_{1,1}^W \rightarrow \text{Aff}$ is the composite of the inclusion $\mathcal{M}_{1,1}^W \hookrightarrow \mathcal{W}$ with $p_{\mathcal{W}}$ (Definition 3.17).

Lean:

[not ready]

Lemma 4.5 (Base-change lifts stay elliptic). *If $(R', W') \in \mathcal{M}_{1,1}^W$ (so $W'.\text{IsElliptic}$) and $\underline{f} : \text{Spec } R \rightarrow \text{Spec } R'$, then the source $(R, W'.\text{map } f)$ of $\lambda_{f, W'}$ (Lemma 3.27) lies in $\mathcal{M}_{1,1}^W$.*

Lean:

[not ready]

Lemma 4.6 (The lifts are strongly cartesian in $\mathcal{M}_{1,1}^W$). *For $(R', W') \in \mathcal{M}_{1,1}^W$ and $\underline{f} : \text{Spec } R \rightarrow \text{Spec } R'$, the arrow $\lambda_{f, W'}$ is strongly cartesian for $p : \mathcal{M}_{1,1}^W \rightarrow \text{Aff}$.*

Lean:

[not ready]

Proof. $\mathcal{M}_{1,1}^W \subseteq \mathcal{W}$ is full, so its hom-sets between elliptic objects coincide with those of $\mathcal{W}$. The universal property (Definition 1.4) for p quantifies over test objects of $\mathcal{M}_{1,1}^W$, a subfamily of those for $p_{\mathcal{W}}$; the unique factorisation produced in Lemma 3.27 has source a test object of $\mathcal{M}_{1,1}^W$ and target $(R, W'.\text{map } f) \in \mathcal{M}_{1,1}^W$ (Lemma 4.5), hence is a morphism of $\mathcal{M}_{1,1}^W$. Thus $\lambda_{f, W'}$ is strongly cartesian in $\mathcal{M}_{1,1}^W$. $\square$

Lemma 4.7 ($p : \mathcal{M}_{1,1}^W \rightarrow \text{Aff}$ is prefibered). *For every $\underline{f} : \text{Spec } R \rightarrow \text{Spec } R'$ and $(R', W') \in \mathcal{M}_{1,1}^W$, the arrow $\lambda_{f, W'}$ is a strongly cartesian lift of $\underline{f}$ in $\mathcal{M}_{1,1}^W$ with target (R', W') .*

Lean:

[not ready]

Lemma 4.8 (The fibres of $\mathcal{M}_{1,1}^W$ are groupoids). *For $R \in \text{Aff}$, $p.\text{Fiber}(\text{Spec } R)$ is the full subcategory of $p_{\mathcal{W}}.\text{Fiber}(\text{Spec } R)$ (Lemma 3.23) on the elliptic curves, and it is a groupoid.*

Lean:

[not ready]

Proof. By Lemma 4.2, if $W.\text{IsElliptic}$ and $C \cdot W = W_0$ then $W_0.\text{IsElliptic}$; the fibre is thus the full subcategory of Lemma 3.23 on elliptic objects, and the inverse C^{-1} (Lemma 3.22) preserves ellipticity, so it is a groupoid. $\square$

Theorem 4.9 ($\mathcal{M}_{1,1}^W$ is a CFG). *$p : \mathcal{M}_{1,1}^W \rightarrow \text{Aff}$ is fibered in groupoids.*

Lean:

[not ready]

Proof. p is prefibered (Lemma 4.7) and its fibres are groupoids (Lemma 4.8); apply Proposition 3.30. $\square$

4.2 The j -invariant

Definition 4.10 (The affine j -line). $\mathbb{A}^1$ is the CFG over Aff represented by $\mathbb{Z}[j]$: its fibre over R is the discrete category on the underlying set of R (objects (R, x) with $x \in R$; only identity morphisms), and base change is application of the ring map.

Lean:

[not ready]

Lemma 4.11 (j is invariant under variable change). For $W.\text{IsElliptic}$ and $C \in \text{VC}(R)$, $(C \cdot W).j = W.j$.

Lean: `WeierstrassCurve.variableChange_j`

[not ready]

Lemma 4.12 (j is natural under base change). For $W.\text{IsElliptic}$ and $f : R \rightarrow A$, $(W.\text{map } f).j = f(W.j)$.

Lean: `WeierstrassCurve.map_j`

[not ready]

Definition 4.13 (The j -line morphism). The assignment $(R, W) \mapsto (R, W.j)$ and $(\underline{f}, C) \mapsto \underline{f}$ defines a morphism of CFGs $j : \mathcal{M}_{1,1}^W \rightarrow \mathbb{A}^1$ over Aff : Lemma 4.11 makes it well-defined on morphisms (the VC -component acts trivially on j), and Lemma 4.12 makes it commute with base change.

Lean:

[not ready]

Lemma 4.14 (ofJ realises a j -invariant). Over a field F , $(\text{ofJ } j_0).j = j_0$ for every $j_0 \in F$.

Lean: `WeierstrassCurve.ofJ_j`

[not ready]

Proposition 4.15 (Fibrewise essential surjectivity of j over a field). Over a field F , j is essentially surjective on the fibre: every $j_0 \in F$ equals $W.j$ for $W = \text{ofJ } j_0$ (Lemma 4.14). The complementary fibrewise injectivity holds over separably closed fields and is the target `isom_iff_j_eq`.

Lean:

[not ready]

4.3 The quotient presentation

Definition 4.16 (Universal Weierstrass ring). $P := \mathbb{Z}[a_1, a_2, a_3, a_4, a_6] = \text{MvPolynomial } (\text{Fin } 5) \mathbb{Z}$, the polynomial ring on the five Weierstrass coefficients.

Lean:

[not ready]

Definition 4.17 (Tautological curve and its discriminant). $W^P : \text{WeierstrassCurve } P$ is the curve whose coefficients are the five generators of P ; set $\Delta_P := W^P.\Delta \in P$.

Lean:

[not ready]

Definition 4.18 (Universal base ring). $A := P[\Delta_P^{-1}] = \text{Localization.Away } \Delta_P$, with structure map $\iota : P \rightarrow A$.

Lean:

[not ready]

Definition 4.19 (Universal elliptic curve). $W^{\text{univ}} := W^P.\text{map } \iota : \text{WeierstrassCurve } A$.

Lean:

[not ready]

Lemma 4.20 (The universal curve is elliptic). $W^{\text{univ}}.\text{IsElliptic}$, because $\iota(\Delta_P) = W^{\text{univ}}.\Delta$ is a unit in $A = \text{Localization.Away } \Delta_P$.

Lean:

[not ready]

Definition 4.21 (The group functor of variable changes). $G : \text{CommRing} \rightarrow \text{Grp}$ is the functor $R \mapsto \text{VC}(R)$, with the group law of Definition 1.15 and functoriality map (Definition 3.1).

Lean:

[not ready]

Lemma 4.22 (Representability of G). G is representable by $\mathbb{Z}[u, u^{-1}, r, s, t]$; as a scheme $G \cong \mathbb{G}_m \times \mathbb{G}_a^3$, with group law the semidirect product $\mathbb{G}_a^3 \rtimes \mathbb{G}_m$ for the evident weights.

Lean:

[not ready]

Definition 4.23 (The G -action on $\text{Spec } A$). G acts on $\text{Spec } A$ by variable change, $C \cdot W^{\text{univ}}$; the co-action is the A -algebra map $A \rightarrow A \otimes \mathbb{Z}[u, u^{-1}, r, s, t]$ dual to the transformed coefficients of Definition 1.15.

Lean:

[not ready]

Lemma 4.24 (Points of $\text{Spec } A$ are elliptic curves). For $R \in \text{CommRing}$, the assignment $(\varphi : A \rightarrow R) \mapsto W^{\text{univ}}.\text{map } \varphi$ is a bijection from $\text{Hom}_{\text{CommRing}}(A, R)$ to the set of elliptic $W : \text{WeierstrassCurve } R$.

Lean:

[not ready]

Proof. By the universal property of $P = \text{MvPolynomial } (\text{Fin } 5) \mathbb{Z}$, a ring map $P \rightarrow R$ is a choice of $(a_i) \in R^5$; by the universal property of $\text{Localization.Away } \Delta_P$ it extends to $A \rightarrow R$ iff the image of Δ_P , namely $W.\Delta$, is a unit, i.e. iff $W.\text{IsElliptic}$ (Definition 1.17). $\square$

Lemma 4.25 (The bijection is G -equivariant and natural). Under Lemma 4.24, the $G(R) = \text{VC}(R)$ -action on $\text{Hom}_{\text{CommRing}}(A, R)$ (Definition 4.23) corresponds to the variable-change action on elliptic curves, and the bijection is natural in R along base change (Lemma 3.6).

Lean:

[not ready]

Theorem 4.26 (Quotient presentation). The universal curve induces an equivalence of CFGs over Aff

$$\mathcal{M}_{1,1}^W \simeq [\text{Spec } A / G],$$

where $[\text{Spec } A / G]$ is the quotient CFG whose fibre over R is the action groupoid $G(R) \ltimes (\text{Spec } A)(R)$.

Lean:

[not ready]

Proof blueprint. On each fibre, Lemma 4.24 identifies the objects $(\mathrm{Spec} A)(R)$ of $[\mathrm{Spec} A/G]$ with the elliptic curves over R , and Lemma 4.25 identifies the $G(R)$ -action with the variable-change action; hence the action groupoid $G(R) \ltimes (\mathrm{Spec} A)(R)$ agrees with the fibre groupoid of $\mathcal{M}_{1,1}^W$ (Lemma 3.23 on elliptic curves). Naturality in R (Lemma 4.25) upgrades these fibre equivalences to an equivalence of CFGs. $\square$

Remark 4.27. Theorem 4.26 exhibits $\mathcal{M}_{1,1}^W$ as a global quotient stack. It is the entry point to the smoothness of $\mathcal{M}_{1,1}$ (the atlas $\mathrm{Spec} A$ is smooth over $\mathbb{Z}$) and to the automorphism group schemes (stabilisers of the G -action), including the generic automorphism $[-1]$ and the extra automorphisms at $j = 0, 1728$.

5 Descent and the stack condition

We now formulate what it means for the CFG $\mathcal{M}_{1,1}^W$ to be a stack for τ_{fppf} , and reduce it to descent statements. As flagged in Warning 1.12, the notion of stack is not in Mathlib; the definitions here are new.

Definition 5.1 (Isom presheaf). For a CFG $p : \mathcal{X} \rightarrow \mathcal{S}$, $S \in \mathcal{S}$ and $x, y \in \mathcal{X}(S)$, the presheaf $\mathrm{Isom}_S(x, y)$ on the slice $\mathcal{S}_{/S}$ sends $(T \xrightarrow{a} S)$ to the set of isomorphisms $a^*x \xrightarrow{\sim} a^*y$ in $\mathcal{X}(T)$, with restriction by further base change.

Lean:

[not ready]

Definition 5.2 (Prestack, stack). Let $(\mathcal{S}, \tau)$ be a site and $p : \mathcal{X} \rightarrow \mathcal{S}$ a CFG.

- (a) p is a *prestack* for τ if for all S and all $x, y \in \mathcal{X}(S)$ the presheaf $\mathrm{Isom}_S(x, y)$ is a τ -sheaf on $\mathcal{S}_{/S}$ (equivalently: morphisms glue).
- (b) A *descent datum* for a covering $\{U_i \rightarrow S\}$ is a family $x_i \in \mathcal{X}(U_i)$ with isomorphisms $\varphi_{ij} : x_i|_{U_{ij}} \xrightarrow{\sim} x_j|_{U_{ij}}$ satisfying the cocycle condition on triple overlaps. It is *effective* if it is isomorphic to the descent datum induced by some $x \in \mathcal{X}(S)$.
- (c) p is a *stack* for τ if it is a prestack and every descent datum for every τ -covering is effective.

Lean:

[not ready]

Proposition 5.3 (Representability of Isom for $\mathcal{M}_{1,1}^W$). For W_1, W_2 elliptic over R , the presheaf $\mathrm{Isom}_{\mathrm{Spec} R}(W_1, W_2)$ is representable by an affine R -scheme: a closed subscheme of $G_R = \mathbb{G}_m \times \mathbb{G}_a^3$ (base-changed to R) cut out by the five equations $C \cdot W_1 = W_2$ of Definition 1.15 (coefficient-wise). In particular it is an fppf sheaf.

Lean:

[not ready]

Proof blueprint. A T -point of $\mathrm{Isom}(W_1, W_2)$ over $a : R \rightarrow T$ is, by Lemma 3.23, a $C \in \mathrm{VC}(T)$ with $C \cdot (W_1.\mathrm{map} a) = W_2.\mathrm{map} a$. Equivalently it is a T -point of $G_R = \mathrm{Spec} R[u, u^{-1}, r, s, t]$ satisfying the five polynomial equations obtained by equating coefficients in $C \cdot W_1 = W_2$ (a system with $\mathbb{Z}$ -coefficients in $u^{\pm 1}, r, s, t$ and the $a_i(W_1), a_i(W_2)$). Hence $\mathrm{Isom}(W_1, W_2) = \mathrm{Spec}(R[u, u^{-1}, r, s, t]/(\text{those 5 relations}))$ an affine scheme over R ; representable functors on Aff are fppf sheaves once faithfully flat descent for morphisms is available (see Lemma 5.4). $\square$

Lemma 5.4 (Representable $\Rightarrow$ fppf sheaf). *The functor of points of any affine scheme is a sheaf for τ_{fppf} on Aff . Equivalently: for a faithfully flat ring map $R \rightarrow R'$ the Amitsur complex*

$$0 \rightarrow R \rightarrow R' \rightrightarrows R' \otimes_R R' \rightarrow \cdots$$

is exact; hence $R \rightarrow R'$ is an equaliser of $R' \rightrightarrows R' \otimes_R R'$ and any R -algebra of finite type has the descent property.

Lean: `RingHom.FaithfullyFlat, Algebra.amitsurComplex (target)`

[not ready]

Status. The exactness of the Amitsur (Čech–Amitsur) complex for a faithfully flat $R \rightarrow R'$ is the key input. Mathlib has `Module.FaithfullyFlat` and the one-step statement that $R \rightarrow R'$ is injective with the expected equaliser description in low degree; the full cosimplicial exactness and the packaging “representable $\Rightarrow$ fppf sheaf” are targets. This lemma is a self-contained prerequisite, independent of elliptic curves. $\square$

Theorem 5.5 ($\mathcal{M}_{1,1}^W$ is an fppf prestack). *$p : \mathcal{M}_{1,1}^W \rightarrow \text{Aff}$ is a prestack for τ_{fppf} .*

Lean:

[not ready]

Proof. Immediate from Proposition 5.3 and Lemma 5.4: each `Isom` presheaf is representable, hence an fppf sheaf, which is precisely the prestack condition of Definition 5.2(a). $\square$

Theorem 5.6 (Effectivity of descent for $\mathcal{M}_{1,1}^W$). *Every fppf descent datum for $\mathcal{M}_{1,1}^W$ is effective; hence $p : \mathcal{M}_{1,1}^W \rightarrow \text{Aff}$ is a stack for τ_{fppf} .*

Lean:

[not ready]

Proof blueprint and prerequisite ledger. Let $\{R \rightarrow R_i\}$ be an fppf cover with $R \rightarrow R' = \prod R_i$ faithfully flat. A descent datum is a curve W' over R' together with an isomorphism $\sigma : \text{pr}_1^* W' \xrightarrow{\sim} \text{pr}_2^* W'$ over $R' \otimes_R R'$ satisfying the cocycle condition over $R' \otimes_R R' \otimes_R R'$. Effectivity means: there is an elliptic W over R and an isomorphism $W' \cong W.\text{map}(R \rightarrow R')$ carrying the canonical datum to (W', σ) . The reduction proceeds by descending the coefficients:

- (i) Each coefficient $a_k(W') \in R'$ together with the transformation prescribed by σ (a variable change $C \in \text{VC}(R' \otimes_R R')$) constitutes a descent datum for the quasi-coherent sheaf underlying the Weierstrass data. Because σ need not be the identity, the naive coefficient-wise descent fails in general; what descends is the pair (line bundle ω , section data), which is why the twist by `Pic` appears (Warning 1.18).
- (ii) **Prerequisite: fppf descent for quasi-coherent modules.** For a faithfully flat $R \rightarrow R'$, the category of R -modules is equivalent to the category of R' -modules with descent data (Grothendieck). This is *not* in Mathlib; it is the load-bearing missing input. Its statement: the base-change functor $\text{Mod}_R \rightarrow \text{Desc}(R'/R)$ is an equivalence.
- (iii) **Prerequisite: fppf descent for the multiplicative and additive group.** $\mathbb{G}_m$ and $\mathbb{G}_a$ are fppf sheaves (special case of (ii) / of Lemma 5.4); this handles the descent of the unit u and of the translation parameters r, s, t in σ .

- (iv) Granting (ii)–(iii), the descent datum (W', σ) is effective: the Hodge line bundle $\omega' = p'_* \Omega$ (in the affine model, the invertible R' -module attached to σ) descends to an invertible R -module ω ; choosing (fppf-locally) a trivialisation recovers a Weierstrass equation, and the cocycle condition guarantees the descended $(a_k) \in R$ define an elliptic curve W over R inducing (W', σ) . When ω is globally trivial (e.g. $\text{Pic}(R) = 0$) the descent is a genuine Weierstrass curve over R ; in general W exists in the stackification.

Thus, modulo the descent prerequisites (ii)–(iii), $\mathcal{M}_{1,1}^W$ satisfies effective descent and is an fppf stack. The honest statement of current feasibility: the prestack theorem 5.5 is reachable from present Mathlib plus Lemma 5.4; the effectivity theorem requires the quasi-coherent descent package (ii), a substantial independent development. $\square$

Remark 5.7 (Stackification). Since Mathlib lacks stackification, the clean statement “the fppf-stackification of $\mathcal{M}_{1,1}^W$ is $\mathcal{M}_{1,1}$ ” (Theorem 6.2) presupposes a new **Stackification** construction (the 2-categorical analogue of `presheafToSheaf`). Building it is a self-contained category-theory target: sheafify the object- and morphism-presheaves and check the universal property. It is listed as milestone **M8**.

6 Comparison with the geometric moduli stack

Definition 6.1 (Elliptic curve over a scheme; $\mathcal{M}_{1,1}$). An *elliptic curve* over a scheme S is a morphism $p : E \rightarrow S$ that is proper, smooth, of relative dimension one, with geometrically connected fibres of arithmetic genus one, together with a section $e : S \rightarrow E$. A morphism of elliptic curves over $a : T \rightarrow S$ is a cartesian square inducing $E_T \xrightarrow{\sim} a^* E$ compatibly with sections. The resulting category $\mathcal{M}_{1,1}$ over Sch (or over Aff , or over $\text{Sch}/\mathbb{Z}$) is fibered in groupoids; this is the *moduli stack of elliptic curves* [2, 4, 5].

Lean:

[not ready]

Theorem 6.2 (Comparison). *There is a morphism of CFGs $\Phi : \mathcal{M}_{1,1}^W \rightarrow \mathcal{M}_{1,1}$ over Aff sending an elliptic Weierstrass curve W over R to Proj of its homogeneous coordinate ring (the projective plane cubic with the point at infinity as section). Then:*

- (a) *For R with $\text{Pic}(R)[12] = 0$ (in particular $\text{Pic}(R) = 0$, e.g. R local or a PID), $\Phi_R : \mathcal{M}_{1,1}^W(R) \rightarrow \mathcal{M}_{1,1}(R)$ is an equivalence of groupoids.*
- (b) *In general Φ exhibits $\mathcal{M}_{1,1}$ as the fppf-stackification of $\mathcal{M}_{1,1}^W$; equivalently, every elliptic curve over S admits a Weierstrass equation Zariski-locally on S , and the obstruction to a global one lies in $\text{Pic}(S)$ via the Hodge bundle.*

Lean:

[not ready]

Discussion and prerequisites. The construction of Φ and the proof of (a),(b) are the content of [6, II.2, III.1, III.3], [1, §2.1–2.2], [4, Ch. 2]. The essential geometric facts are: (1) an elliptic curve E/S has $p_* \mathcal{O}_E = \mathcal{O}_S$ and an invertible $\omega = p_* \Omega_{E/S}$; (2) Riemann–Roch for the section divisor yields a closed immersion $E \hookrightarrow \mathbb{P}(p_* \mathcal{O}_E(3e))$ realising E as a Weierstrass cubic once ω is trivialised; (3) two Weierstrass models of the same E differ by a variable change, matching Definition 1.15. Each of these presupposes machinery not in Mathlib.

Missing scheme-theoretic prerequisites (future-work ledger). In rough dependency order:

- (P1) proper morphisms of schemes and their stability (`AlgebraicGeometry.IsProper`, partial);
- (P2) smooth morphisms and relative dimension (partial in `AlgebraicGeometry.Morphisms.Smooth`);
- (P3) relative differentials $\Omega_{E/S}$ and the dualising/Hodge line bundle;
- (P4) higher direct images $R^i p_*$ and cohomology-and-base-change, giving $p_* \mathcal{O}_E = \mathcal{O}_S$ and invertibility of ω ;
- (P5) relative Riemann–Roch / relative Proj and the very-ampleness of $\mathcal{O}_E(3e)$;
- (P6) the classification of Weierstrass models of a fixed E (the “ $3e$ ” embedding and its normal form), reproducing Definition 1.15;
- (P7) faithfully flat / fppf descent for quasi-coherent sheaves (Theorem 5.6(ii)) and for schemes;
- (P8) the 2-categorical stackification of Remark 5.7.

Consequently, within a horizon fixed by present Mathlib, the achievable target is Parts 3–4 (the CFG $\mathcal{M}_{1,1}^W$, its groupoid fibres, and the quotient presentation) together with Theorem 5.5 (the fppf prestack property, modulo Lemma 5.4). Theorem 6.2 and Theorem 5.6 are longer-term and are stated here to fix the mathematically correct target and to make the dependency on (P1)–(P8) explicit. $\square$

Remark 6.3 (Algebraicity of $\mathcal{M}_{1,1}$). With (P1)–(P8) in hand, Theorem 4.26 shows $\mathcal{M}_{1,1}$ is a global quotient stack $[\mathrm{Spec} A/G]$, hence a smooth algebraic (Artin) stack of relative dimension one over $\mathrm{Spec} \mathbb{Z}$, separated, and Deligne–Mumford over $\mathbb{Z}[1/6]$ (the automorphism group schemes are finite étale away from characteristics 2, 3) [2, 5]. These are statements about $\mathcal{M}_{1,1}$, downstream of the entire ledger, recorded for orientation.

7 Milestone ladder

The following orders the work by Mathlib-readiness. Milestones **M0**–**M5** are reachable from present Mathlib; **M6**–**M9** depend on independent developments (descent, stackification, scheme geometry).

M0 — Substrate audit.

Confirm the interface of Definitions 1.3–1.11, 1.13–1.17 against the pinned Mathlib commit; add missing thin API (`VariableChange.map`, `map_j`, `IsElliptic.map` if absent). *Depends on:* nothing.

M1 — Equivariance.

Definition 3.1; Lemmas 3.2, 3.3, 3.4, 3.5; Lemma 3.6. *Depends on:* M0.

M2 — The fibered category $\mathcal{W}$.

Definitions 3.7–3.12 with Lemmas 3.9, 3.11; the category axioms 3.13–3.15 and Definition 3.16; the projection 3.17; the lift/fibre lemmas 3.18–3.23; strong cartesianness 3.24–3.27; prefiberedness 3.28; Theorem 3.29; the groupoid criterion 3.30–3.32; Corollary 3.33. *Depends on:* M1 and the `FiberedCategory` API.

M3 — The moduli CFG $\mathcal{M}_{1,1}^W$.

Lemmas 4.1, 4.2; Definitions 4.3, 4.4; Lemmas 4.5–4.8; Theorem 4.9; the j -invariant nodes 4.10, 4.11, 4.12, 4.13, 4.14, 4.15. *Depends on:* M2.

M4 — Quotient presentation.

Definitions 4.16–4.19 with Lemma 4.20; the group nodes 4.21, 4.22, 4.23; Lemmas 4.24, 4.25; Theorem 4.26. *Depends on:* M3 and the functor-of-points/Yoneda API for `MvPolynomial` and `Localization.Away`.

M5 — Pseudofunctor route (optional).

Definition 3.34; Constructions 3.35, 3.36; along it Theorem 3.29 is discharged by Proposition 1.10. *Depends on:* M1 (to build the pseudofunctor $\mathbf{F}$, including its coherences) and `Pseudofunctor.CoGrothendieck` together with `FiberedCategory.Grothendieck` (`CoGrothendieck.inst`).

M6 — fppf site.

Definition 2.1; Lemma 2.2; Lemma 5.4. *Depends on:* flatness/finite-presentation base-change API and the Amitsur complex.

M7 — Prestack.

Definitions 5.1, 5.2; Proposition 5.3; Theorem 5.5. *Depends on:* M3, M6.

M8 — Descent and stackification.

Theorem 5.6 (needs quasi-coherent fppf descent, P7); Remark 5.7 (the stackification functor). *Depends on:* M7 and P7.

M9 — Geometric comparison.

Definition 6.1; Theorem 6.2; Remark 6.3. *Depends on:* M8 and the scheme-geometry ledger (P1)–(P6).

Appendix A. Explicit formulas and their Mathlib names

The variable-change action is that of Definition 1.15, matching [6, III, Table 3.1] and the Mathlib instance `WeierstrassCurve.instMulActionVariableChange`. The transformation weights are: $c_4 \mapsto u^{-4}c_4$, $c_6 \mapsto u^{-6}c_6$, $\Delta \mapsto u^{-12}\Delta$, and $j \mapsto j$ (invariant); the corresponding Mathlib lemmas are `variableChange_c4`, `variableChange_c6`, `variableChange_Delta`, `variableChange_j`. Under base change $f : R \rightarrow A$ the invariants are natural: c_4, c_6, Δ, j commute with f , via `map_c4`, `map_c6`, `map_Delta`, `map_j`. The identity $c_4^3 - c_6^2 = 1728\Delta$ is `WeierstrassCurve.c4_cubed_...` (coefficient identity in `Mathlib.AlgebraicGeometry.EllipticCurve.Weierstrass`).

Appendix B. Formalisation status by item

Already in Mathlib (mark [\[in Mathlib / formalised\]](#)): the fibered-category API (Definitions 1.3–1.7, Lemma 1.5, Definition 1.6); the Grothendieck and CoGrothendieck constructions of a pseudofunctor (Remark 1.9) and the theorem that the CoGrothendieck projection is a fibered category (Proposition 1.10, `Mathlib.CategoryTheory.FiberedCategory.Grothendieck`); sites and sheaves (Definition 1.11); the Weierstrass/elliptic-curve API (Definitions 1.13–1.17, Lemmas 1.14, 1.16).

New but reachable now (M1–M5, M6-partial): Lemma 3.6; the category $\mathcal{W}$ and its CFG structure; $\mathcal{M}_{1,1}^W$ and Theorem 4.9; the quotient presentation Theorem 4.26; the fppf pretopology and Lemma 5.4 (the latter needing the Amitsur complex).

New and dependent on major developments (M8–M9): Theorem 5.6 (quasi-coherent fppf descent, P7); stackification (Remark 5.7); Definition 6.1 and Theorem 6.2 (the scheme-geometry ledger P1–P8); Remark 6.3.

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